DS HOMEWORK 3

1.Write a C program to store n elements in an array using dynamic memory allocation and find largest number.

Code:

```c
#include <stdio.h>
#include <stdlib.h>
int main(){
    int n,*arr,i,max;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    arr=(int *)malloc(n*sizeof(int));
    max=arr[0];
    printf("Enter the values:");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
        if(max<arr[i]){
            max=arr[i];
        }
    }
    printf("The elements in the array:\n");
    for(i=0;i<n;i++){
        printf("%d ",arr[i]);
    }
    printf("\nThe largest element in an array is %d",max);
    free(arr);
    return 0;
}
```

Output:

```
PS D:\ds_hw3> ./1
Enter the number of elements:3
Enter the values:23
32
12
The elements in the array:
23 32 12
The largest element in an array is 32
PS D:\ds_hw3>
```

2.Print an array in reverse order using pointers.

Code:

```c
#include <stdio.h>
int main(){
    int *p,arr[10],n,i;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    printf("Enter the values:");
```

```c
for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    p=arr;
    printf("Array in reverse order:\n");
    for(i=n-1;i>=0;i--){
        printf("%d ",*(p+i));
    }
    return 0;
}
```

Output:

```
PS D:\ds_hw3> gcc 2.c -o 2
PS D:\ds_hw3> ./2
Enter the number of elements:4
Enter the values:2
12
32
45
Array in reverse order:
45 32 12 2
PS D:\ds_hw3>
```

3.Write a C program to store information of student using structure. Sort them by Name.
Code:

```c
#include <stdio.h>
#include <string.h>
struct student{
    char name[20];
    int rollno;
};
int main(){
    struct student s[10];
    int n,i,j;
    printf("Enter the number of students:");
    scanf("%d",&n);
    printf("Enter details:");
    for(i=0;i<n;i++){
        scanf("%s",s[i].name);
        scanf("%d",&s[i].rollno);
    }
    for(i=0;i<n-1;i++){
        for(j=i;j<n;j++){
            if(strcmp(s[i].name,s[j].name)>0){
                char temp[20];
                int tem;
                strcpy(temp,s[i].name);
```

```
                strcpy(s[i].name,s[j].name);
                 strcpy(s[j].name,temp);
                 tem=s[i].rollno;
                 s[i].rollno=s[j].rollno;
                 s[j].rollno=tem;
             }
         }
    }
    printf("Names in sorted manner:\n");
    for(i=0;i<n;i++){
        printf("Name: %s\n",s[i].name);
        printf("Rollno: %d\n",s[i].rollno);
    }
    return 0;
}
```

Output:

```
PS D:\ds_hw3> gcc 3.c -o 3
PS D:\ds_hw3> ./3
Enter the number of students:2
Enter details:Shobiu
56
Nagul
45
Names in sorted manner:
Name: Nagul
Rollno: 45
Name: Shobiu
Rollno: 56
```

4.Write a C program to add 'n' distances given in inch-feet system using structure.

Code:

```
#include <stdio.h>
struct distance{
    int feet;
    int inches;
};
int main(){
    struct distance d,temp;
    int n,r,c,i;
    d.feet=0;
    d.inches=0;
    printf("Enter number of distances: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++) {
        printf("Enter distance %d: ", i + 1);
        scanf("%d %d", &temp.feet, &temp.inches);
        d.feet += temp.feet;
        d.inches += temp.inches;
```

```
 }
    if(d.inches>=12){
        n=d.inches/12;
        r=d.inches%12;
        d.feet+=n;
        d.inches=r;
    }
    printf("Distance:\n");
    printf("Feet: %d\nInches: %d",d.feet,d.inches);
    return 0;
}
```

Output:

```
PS D:\ds_hw3> ./4
Enter number of distances: 2
Enter distance 1: 34
54
Enter distance 2: 56
89
Distance:
Feet: 101
Inches: 11
PS D:\ds_hw3>
```

5.Write a C program to add two complex numbers by passing structure to a function
Code:

```
#include <stdio.h>
struct complex{
    int real;
    int img;
};
struct complex add(struct complex a,struct complex b){
    struct complex temp;
    temp.real=a.real+b.real;
    temp.img=a.img+b.img;
    return temp;
}
int main(){
    struct complex a,b,c;
    printf("Enter two complex numbers:");
    scanf("%d %d %d %d",&a.real,&a.img,&b.real,&b.img);
    c=add(a,b);
    printf("Addition of two complex numbers: %d+%di",c.real,c.img);
    return 0;
}
```

Output:

```
PS D:\ds_hw3> gcc 5.c -o 5
PS D:\ds_hw3> ./5
Enter two complex numbers:34
56
78
4
Addition of two complex numbers: 112+60i
PS D:\ds_hw3>
```

6.Write a C program to maintain inventory management system using structure.

Code:

```c
#include <stdio.h>
struct item{
    int id;
    char name[50];
    int quantity;
    float price;
};
int main(){
    struct item s[10];
    int n,m,i=0,j;
    while(1){
    printf("--Menu--\n1. Add Product\n2. Display Products\n3. Search Product\n4.
Exit\nEnter option:");
    scanf("%d",&n);
    if(n==4){
        break;
    }
    switch(n){
        case 1:
        printf("Enter the product id,name,quantity,price:");
        scanf("%d %s %d %f",&s[i].id,s[i].name,&s[i].quantity,&s[i].price);
        i++;
        break;
        case 2:
        for(j=0;j<i;j++){
            printf("Id:%d\nName:%s
\nQuantity:%d\nPrice:%.2f\n",s[j].id,s[j].name,s[j].quantity,s[j].price);
        }
        break;
        case 3:
        int se,value=0;
        printf("Enter product id: ");
        scanf("%d",&se);
        for(j=0;j<i;j++){
            if(s[j].id==se){
                value=j+1;
```

```
            }
        }
        if(value==0){
            printf("Item not found\n");
        }
        else{
            printf("Item found at %d position\n",value);
        }
        break;
        default:
        printf("Invaild input");
        break;
    }
}
return 0;
}
```

Output:

```
PS D:\ds_hw3> gcc 6.c -o 6
PS D:\ds_hw3> ./6
--Menu--
1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter option:1
Enter the product id,name,quantity,price:12
Milk
3
100
--Menu--
1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter option:1
Enter the product id,name,quantity,price:13
Soda
3
90
--Menu--
1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter option:2
Id:12
Name:Milk
Quantity:3
Price:100.00
Id:13
Name:Soda
```

Quantity:3
Price:90.00
--Menu--
1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter option:3
Enter product id: 12
Item found at 1 position
--Menu--
1. Add Product
2. Display Products
3. Search Product
4. Exit
Enter option:4
PS D:\ds_hw3>

7.Write a C program for decimal to binary, octal and hexadecimal conversion.
Code:

```
#include <stdio.h>
void binary(int n){
    int i,j,arr[10];
        for(i=0;n>=2;i++){
            arr[i]=n%2;
            n=n/2;
        }
        arr[i]=n;
        printf("\nDecimal to Binary:\n");
        for(j=i;j>=0;j--){
            printf("%d ",arr[j]);
        }
}
void octal(int n){
    int i,j,arr[10];
        for(i=0;n>=8;i++){
            arr[i]=n%8;
            n=n/8;
        }
        arr[i]=n;
        printf("\nDecimal to Octal:\n");
        for(j=i;j>=0;j--){
            printf("%d ",arr[j]);
        }
}
void hexadecimal(int n){
    int i,j,arr[10];
        for(i=0;n>=16;i++){
            arr[i]=n%16;
```

```
            n=n/16;
             if(arr[i]>=10 && arr[i]<16){
                 arr[i]=arr[i]-10+'A';
             }
        }
        arr[i]=n;
        if(arr[i]>=10 && arr[i]<16){
        arr[i]=arr[i]-10+'A';}
        printf("\nDecimal to Hexadecimal:\n");
        for(j=i;j>=0;j--){
            if(arr[j]<=9){
            printf("%d ",arr[j]);}
            else{
                printf("%c ",arr[j]);
            }
        }
}
int main(){
    int n;
    printf("Enter the number:");
    scanf("%d",&n);
    binary(n);
    octal(n);
    hexadecimal(n);
}
```

Output:

```
PS D:\ds_hw3> gcc 7.c -o 7
PS D:\ds_hw3> ./7
Enter the number:15

Decimal to Binary:
1 1 1 1
Decimal to Octal:
1 7
Decimal to Hexadecimal:
F
PS D:\ds_hw3>
```

8.Write a C program to calculate difference between two time periods.

Code:

```
#include <stdio.h>
struct time{
    int hour;
    int min;
};
int main(){
    struct time t1,t2,t3;
    printf("Enter the two time periods: ");
```

```
    scanf("%d %d %d %d",&t1.hour,&t1.min,&t2.hour,&t2.min);
    if(t1.min>=60 || t2.min>=\60){
        printf("Invalid input");
        return 0;
    }
    t3.hour=t1.hour-t2.hour;
    t3.min=t1.min-t2.min;
    if(t3.min<0){
        t3.hour-=1;
        t3.min=60+t3.min;
    }
    printf("Difference between time periods:%d hours %d min",t3.hour,t3.min);
    return 0;
}
```

Output:

```
PS D:\ds_hw3> gcc 8.c -o 8
PS D:\ds_hw3> ./8
Enter the two time periods: 45
45
34
56
Difference between time periods:10 hours 49 min
PS D:\ds_hw3>
```

9.Write a C program to create an employee management system using structures.

Code:

```
#include <stdio.h>

struct employee {

    int id;

    char name[50];

    int age;

    char dept[20];

    float salary;

};

int main() {

    struct employee s[10]; int n, m, i = 0, j;

    while (1) {
```

```
  printf("--Menu--\n1. Add \n2. Display \n3. Search \n4. Exit\nEnter option:");

        scanf("%d", &n);

        if (n == 4) {

            break;

        }

        switch (n) {

            case 1:

                printf("Enter the employee id,name,age,dept,salary:");

                scanf("%d %s %d %s %f",
&s[i].id,s[i].name,&s[i].age,s[i].dept,&s[i].salary);

                i++;

                break;

            case 2:

                for (j = 0; j < i; j++) {

                    printf("Id:%d\nName:%s\nAge:%d\nDept:%s\nSalary:%.2f\n",

                            s[j].id,s[j].name,s[j].age,s[j].dept,s[j].salary);

                }

                break;

            case 3: {

                int se, value = 0;

                printf("Enter Employee id: ");

                scanf("%d", &se);

                for (j = 0; j < i; j++) {

                    if (s[j].id == se) {

                        value = j + 1;

                    }
```

```
                }
                if (value == 0) {
                    printf("Employee not found\n");
                }
                else {
                    printf("Employee found at %d position\n", value);
                }
                break;
            }
            default:
                printf("Invalid input\n");
                break;
        }
    }
    return 0;
}
```

Output:

--Menu--

1. Add

2. Display

3. Search

4. Exit

Enter option: 1

Enter the employee id,name,age,dept,salary: 101 Rahul 25 IT 45000

--Menu--

1. Add

2. Display

3. Search

4. Exit

Enter option: 1

Enter the employee id,name,age,dept,salary: 102 Priya 23 CSE 50000

--Menu--

1. Add

2. Display

3. Search

4. Exit

Enter option: 2

Id:101

Name:Rahul

Age:25

Dept:IT

Salary:45000.00

Id:102

Name:Priya

Age:23

Dept:CSE

Salary:50000.00

--Menu--

1. Add

2. Display

3. Search

4. Exit

Enter option: 3

Enter Employee id: 102

Employee found at 2 position

--Menu--

1. Add

2. Display

3. Search

4. Exit

Enter option: 3

Enter Employee id: 105

Employee not found

--Menu--

1. Add

2. Display

3. Search

4. Exit

Enter option: 4

10.Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

Code:

```c
#include <stdio.h>

float arear(float a, float b){

    return a*b;

}

float areat(float b,float h){

    return 0.5*(h*b);

}

float perir(float a,float b,float c){

    return 2*(a+b);

}
```

```c
float perit(float a,float b,float c){

    return a+b+c;

}

int main(){

    float a,b,c,d,e,(*pptr)(float,float,float),(*ptr)(float,float);

    printf("Enter the length and breadth of rectangle:");

    scanf("%f %f",&a,&b);

    printf("Enter the breadth,height and another side of triangle");

    scanf("%f %f %f",&c,&d,&e);

    ptr=arear;

    printf("Area of rectangle: %.2f\n",ptr(a,b));

    pptr=perir;

    printf("Perimeter of rectangle: %.2f\n",pptr(a,b,0));

    ptr=areat;

    printf("Area of triangle: %.2f\n",ptr(c,d));

    pptr=perit;

    printf("Perimeter of triangle: %.2f\n",pptr(c,d,e));

    return 0;

}
```

Output:

```
PS D:\ds_hw3> gcc 10.c -o 10
PS D:\ds_hw3> ./10
Enter the length and breadth of rectangle:2
3
Enter the breadth,height and another side of triangle1
2
3
Area of rectangle: 6.00
Perimeter of rectangle: 10.00
Area of triangle: 1.00
Perimeter of triangle: 6.00
PS D:\ds_hw3>
```